

The Crew Health and Performance Exploration Analog (CHAPEA) is a series of missions conducted by NASA simulating missions on Mars. It consists of three missions, the first of which began on June 25, 2023. The mission is contained in a hangar at the Johnson Space Center in Houston, Texas. Efforts to mimic a real Mars mission include a 22-minute delay in communications, resources limitations, simulated equipment failures, and simulated spacewalks.

Habitat

The habitat, known as Mars Dune Alpha, is a 1700 square foot 3D-printed area designed to simulate the type of structure that would be built on a mission to Mars. It includes crew quarters, an exercise area, a work room, a recreation area, and a crop area. The crop area has been used to grow peppers, tomatoes, and other vegetables. The exterior of the habitat simulates the surface of Mars complete with a backdrop of cliffs and a ground made of red soil, which on a real mission to Mars could be used to 3D print the habitat. To leave the hangar out to the exterior, crew members go through an airlock.

Missions

The CHAPEA project consists of three missions.

CHAPEA 1

NASA began looking for applicants for CHAPEA 1 in 2023 with the following qualifications:

Be either a permanent resident or a citizen of the United States

Be between 30 and 55 years old

Have either 1,000 hours of pilot-in-command time on a jet aircraft or have two years of experience in a related STEM field

Pass the NASA long-duration flight astronaut physical exam

Have a master's degree in STEM, two years of work toward a doctoral program in a STEM field, a M.D. or D.O. degree, or have completed a pilot school program.

The first CHAPEA mission began on June 25, 2023, with four crew members: Kelly Haston, a research scientist and the mission's commander, Ross Brockwell, a structural engineer and public works administrator, Nathan Jones, an emergency medicine physician, and Anca Selariu, a microbiologist. The mission concluded July 6, 2024, with a total duration time of 378 days.

CHAPEA 2

The application process for the second CHAPEA mission was open from February 16 to April 2, 2024. On September 05, 2025 NASA announced the crew selected for the mission, which is expected to begin October 19, 2025 and end on October 31, 2026.

CHAPEA 3

The third and final scheduled CHAPEA mission is scheduled to begin at some point in 2026.

See also

MARS-500

== References ==